23. Scheduling: cron and systemd timers

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Phase: Phase 7 — Services and boot
Estimated time: 1–2 days
Slug: `23-cron-timers`

Learning objectives

- Write crontab entries
- Prefer systemd timers for modern systems
- Handle environment differences in scheduled jobs
- Log scheduled task output

Prerequisites

Complete prior numbered lessons. Prefer a real Linux VM/container for Linux-specific topics.

```
crontab -e
# m h dom mon dow command
0 3 * * * /usr/local/bin/backup.sh
```

systemd timers

More observable than cron on systemd distros — pair `.timer` + `.service`.

Scheduled jobs often miss env vars available interactively — use absolute paths and explicit `Environment=`.

Common mistakes

- Relative paths in cron
- Emailing root floods from cron noise
- Duplicate cron+timer doing same job

Exercises

Work in `exercises/23-cron-timers/`. Run:

```
./test.sh
```

- **A (Drill):** make `./test.sh` pass
- **B (Apply):** note one real command you ran on a Linux host related to this lesson
- **C (Challenge):** rewrite your solution from memory the next day

Do not open `solutions/23-cron-timers/` until you have an honest failing attempt.

Self-check

1. Cron five fields meaning?

2. Why absolute paths?
3. Timer vs cron advantage?

Further reading

- https://man7.org/linux/man-pages/man5/crontab.5.html